

Energy & Momentum

Newton mechanics :

- momentum conservation
- energy conservation

Generalize to relativity?

Momentum in relativity:

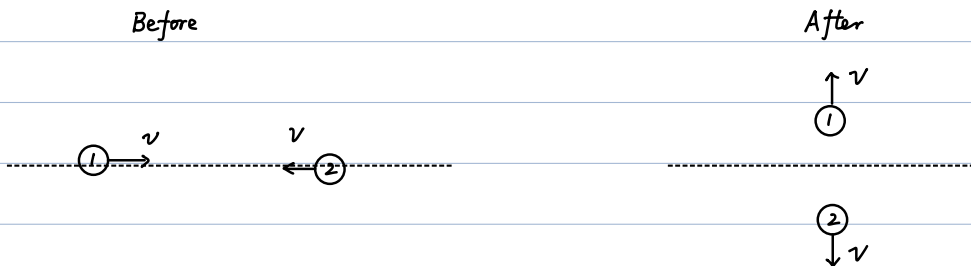
$$\vec{p} = \frac{m_0}{\sqrt{1 - v^2/c^2}} \vec{v}$$

m_0 : mass at rest

check: should satisfy momentum conservation for closed & isolated system

Example: Elastic collision of identical balls

In K' frame (velocity v):



Momentum conservation in K' frame:

$$p'_1 + p'_2 = 0 \quad (\text{before \& after})$$

In K frame:



Velocity transformation:

$$\begin{cases} u_x = \frac{u'_x + v}{1 + \frac{u'_x}{c} \frac{v}{c}} \\ u_y = \frac{u'_y}{\gamma(1 + \frac{u'_x}{c} \frac{v}{c})} \\ u_z = \frac{u'_z}{\gamma(1 + \frac{u'_x}{c} \frac{v}{c})} \end{cases}$$

check velocity after collision:

$$\begin{cases} u_{x1} = \frac{0 + v}{1 + 0} = v \\ u_{y1} = \frac{v}{\gamma(1 + 0)} = \gamma^{-1} v = \sqrt{1 - v^2/c^2} v \end{cases}$$

Momentum in K frame (Non-relativistic version)

before collision:

(b: before)

$$p_{\text{tot}}^{(b)} = m_0 w = \frac{2m_0 v}{1 + v^2/c^2}$$

after collision:

(a: after)

$$p_{\text{tot}}^{(a)} = 2m_0 v \neq p_{\text{tot}}^{(b)}$$

Momentum conservation broken! 

Momentum in K frame (Relativistic version)

$$\begin{aligned} p_{\text{tot}}^{(b)} &= \frac{m_0}{\sqrt{1 - w^2/c^2}} w \\ &= \frac{m_0}{\sqrt{1 - \left(\frac{2v}{1 + v^2/c^2}\right)^2/c^2}} \cdot \frac{2v}{1 + v^2/c^2} \end{aligned}$$

$$= \frac{2m_0 v}{\sqrt{(1 + v^2/c^2)^2 - 4v^2/c^2}}$$

$$= \frac{2m_0 v}{1 - v^2/c^2}$$

$$p_{\text{tot}}^{(a)} = \frac{m_0}{\sqrt{1 - v^2/c^2}} v + \frac{m_0}{\sqrt{1 - v^2/c^2}} v$$

$$= \frac{2m_0 v}{\sqrt{1 - v^2(2 - v^2/c^2)/c^2}}$$

$$= \frac{2m_0 v}{1 - v^2/c^2}$$

$$\Rightarrow p_{\text{tot}}^{(b)} = p_{\text{tot}}^{(a)} \quad \text{momentum conserved!} $$

Note: for $v \ll c \Rightarrow \vec{p} = m_0 \vec{v}$ (recovers non-relativistic) ②

Force (through impulse-momentum):

$$\vec{F} = \frac{d}{dt} \left(\frac{m_0}{\sqrt{1-v^2/c^2}} \vec{v} \right)$$

Notes: 1° $\vec{F}$ changes in different frames.

2° $\vec{F} = m\vec{a}$ NO LONGER holds.

Work-kinetic energy theorem.

$$W = E_b - E_a$$

$$W = \int_a^b \vec{F} \cdot d\vec{r}$$

$$= \int_a^b \frac{d\vec{p}}{dt} \cdot d\vec{r} = \int_a^b d\vec{p} \cdot \vec{v}$$

$$= \int_a^b \frac{\vec{p}}{m} \cdot d\vec{p}$$

$$m \equiv \frac{m_0}{\sqrt{1-(v/c)^2}}$$

$$\text{Since } \begin{cases} \vec{p} = m\vec{v} \\ m = \frac{m_0}{\sqrt{1-(v/c)^2}} \end{cases}$$

$$\Rightarrow m^2 c^2 - p^2 = m_0^2 c^2 \quad (*)$$

$$\Rightarrow m c^2 dm = \vec{p} \cdot d\vec{p}$$

$$\Rightarrow W = \int_a^b c^2 dm$$

$$\begin{cases} a: \text{velocity} = 0, & m = m_0 \\ b: \text{velocity} = v, & m = m \end{cases}$$

$$\Rightarrow W = m c^2 - m_0 c^2 = E_k \text{ (kinetic energy)}$$

↓
Work-kinetic energy theorem

Define

$$\text{total energy: } E = \frac{m_0}{\sqrt{1-(v/c)^2}} c^2$$

$$\text{rest energy: } m_0 c^2$$

$$\Rightarrow \text{Kinetic energy: } E_k = m c^2 - m_0 c^2$$

Note: when $v \ll c$

$$E_k \approx m_0 c^2 \left[1 + \frac{1}{2} \left(\frac{v}{c} \right)^2 \right] - m_0 c^2$$

$$= \frac{1}{2} m_0 v^2 \text{ (recovers non-relativistic)}$$

(3)

Unify $\vec{p}$ & E with mass m :

$$\boxed{\begin{aligned} m &\equiv \frac{m_0}{\sqrt{1-(v/c)^2}} \\ \vec{p} &= m \vec{v} \\ E &= mc^2 \end{aligned}}$$

4-momentum:

$$\vec{P} \equiv \left(\frac{E}{c}, p_x, p_y, p_z \right)$$

$$\left(\frac{E}{c} \right)^2 - \vec{P}^2: \text{ Lorentz invariant (according to *)}$$

4-vector

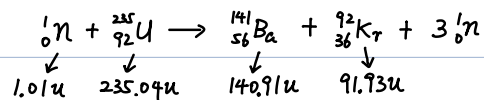
$$(ct, x, y, z):$$

$$s^2 \equiv (ct)^2 - x^2 - y^2 - z^2: \text{ Lorentz invariant}$$

More on $E = mc^2$:

change of mass $\leftrightarrow$ change of energy

Nuclear fission:



$$1\text{u} \approx 1.66 \times 10^{-27} \text{kg}$$

$$\text{mass reduction: } 140.91 + 91.93 + 2.02 - 235.04$$

$$\approx -0.18 \text{ u}$$

$$\approx 3 \times 10^{-28} \text{ kg}$$

Each mole of ${}^{235}\text{U}$ ($\sim 235\text{g}$):

$$3 \times 10^{-28} \text{ kg} \times (3 \times 10^8 \text{ m/s})^2 \times 6.02 \times 10^{23}$$

$$\approx 1.6 \times 10^{13} \text{ J}$$

$$= \frac{1.6 \times 10^{13}}{3.6 \times 10^6} \text{ kWh} \approx 4.4 \times 10^6 \text{ kWh}$$

$$1 \text{ kWh} \sim 0.5 \text{ RMB}$$

$$\sim 2.2 \times 10^6 \text{ RMB}$$

chain reaction: ${}^1_0\text{n} \rightarrow 3 {}^1_0\text{n} \rightarrow 9 {}^1_0\text{n} \rightarrow \dots$

Twin paradox

{ Frank (stays at earth)

{ Mary (flies to α -Centauri, ~ 4 light years away, $v = 0.8c$)

半人马座

$\Rightarrow$ 5 year per one-way trip

(light year: ly)

The paradox:

• In Earth frame (Frank):

After 10 years, Mary is back. Frank aged 10 years.

Mary's clock is slowed down, Mary aged $\sqrt{1-0.8^2} \times 10 = 6$ years.

• In spacecraft frame (Mary)

Mary see distance contracted: $L = \sqrt{1-0.8^2} \times 4 = 2.4$ light years,

takes 6 years for roundtrip.

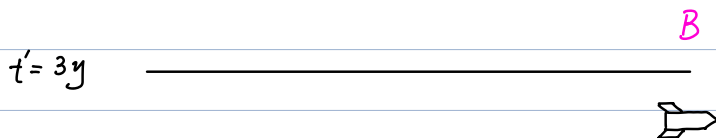
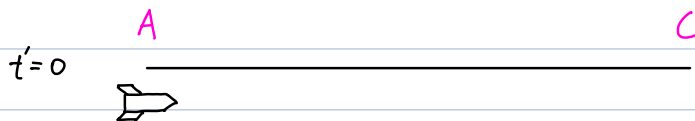
Frank's clock slowed down, Frank aged $\sqrt{1-0.8^2} \times 6 = 3.6$ years.

Note: spacecraft acceleration $\rightarrow$ in principle non-inertial frame.

In the following, we assume negligible acceleration time.

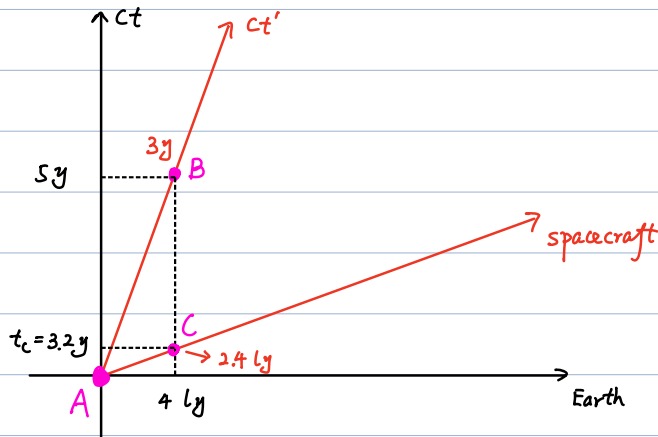
Illustrate using half of the roundtrip.

Spacecraft frame



$$C: \begin{pmatrix} ct_c \\ x_c \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} ct'_c \\ x'_c \end{pmatrix} \xrightarrow{0} = \begin{pmatrix} \beta\gamma x'_c \\ \gamma x'_c \end{pmatrix} = \begin{pmatrix} 3.2 \text{ ly} \\ 4 \text{ ly} \end{pmatrix} \quad \begin{matrix} \beta = 0.8 \\ \gamma = \frac{5}{3} \end{matrix}$$

$\Rightarrow t_c = 3.2 y$ $\xrightarrow{2.4 \text{ ly}}$ ⑤



Mary feels that Frank's clock slowed down:

$$t_B - t_C = 5y - 3.2y = 1.8y = \sqrt{1 - 0.8^2} \times 3y \quad \checkmark$$

So, Frank's clock is slowed down, but the time interval is $C \rightarrow B$.

Finally,

- After 10 y in Earth frame, Mary is back
- Frank aged 10, Mary aged 6.

Doppler effect of light

$$f = \frac{c}{\lambda}$$

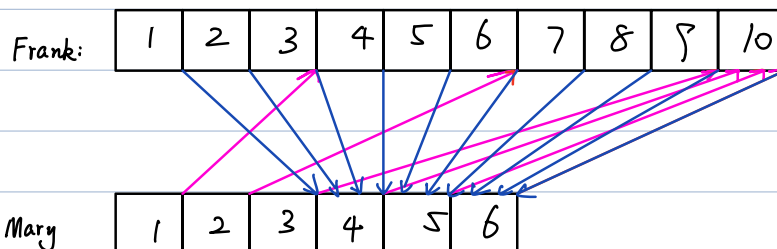
$$f = \sqrt{\frac{1 + v/c}{1 - v/c}} f_0$$

(proof omitted)

Each year, Frank & Mary send one radio signal ($f_0 = 1 \text{ y}^{-1}$)

First half trip: $f = \sqrt{\frac{1 - 0.8}{1 + 0.8}} f_0 = \frac{1}{3} \text{ y}^{-1}$ (3 year per signal)

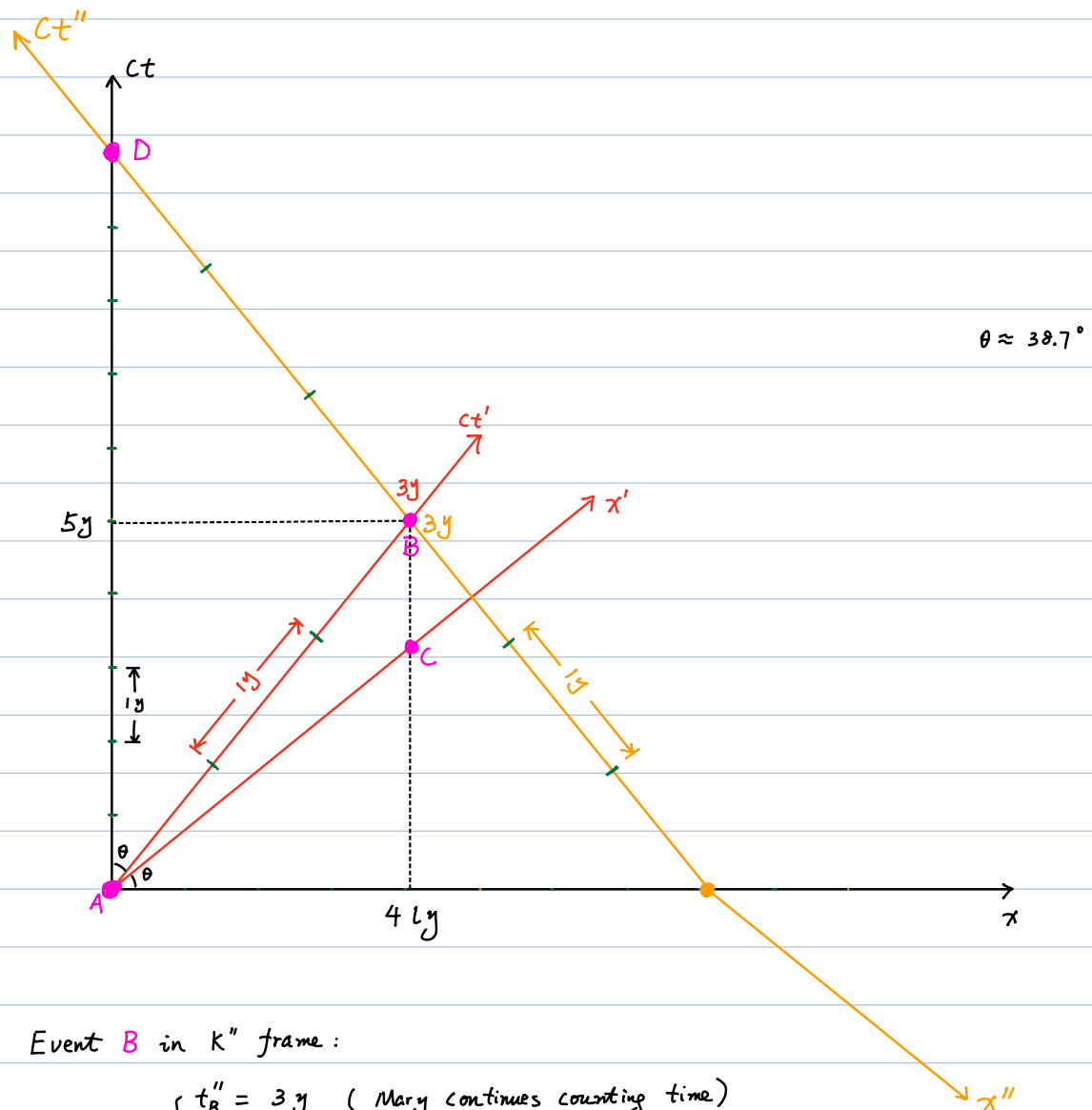
Second half: $f = \sqrt{\frac{1 + 0.8}{1 - 0.8}} f_0 = 3 \text{ y}^{-1}$ (3 signals per year)



K frame : Earth (Frank)

K' frame: space craft to α -centauri (Mary) $\xrightarrow{v}$

K'' frame: spacecraft to earth (Mary) $\xleftarrow{-v}$



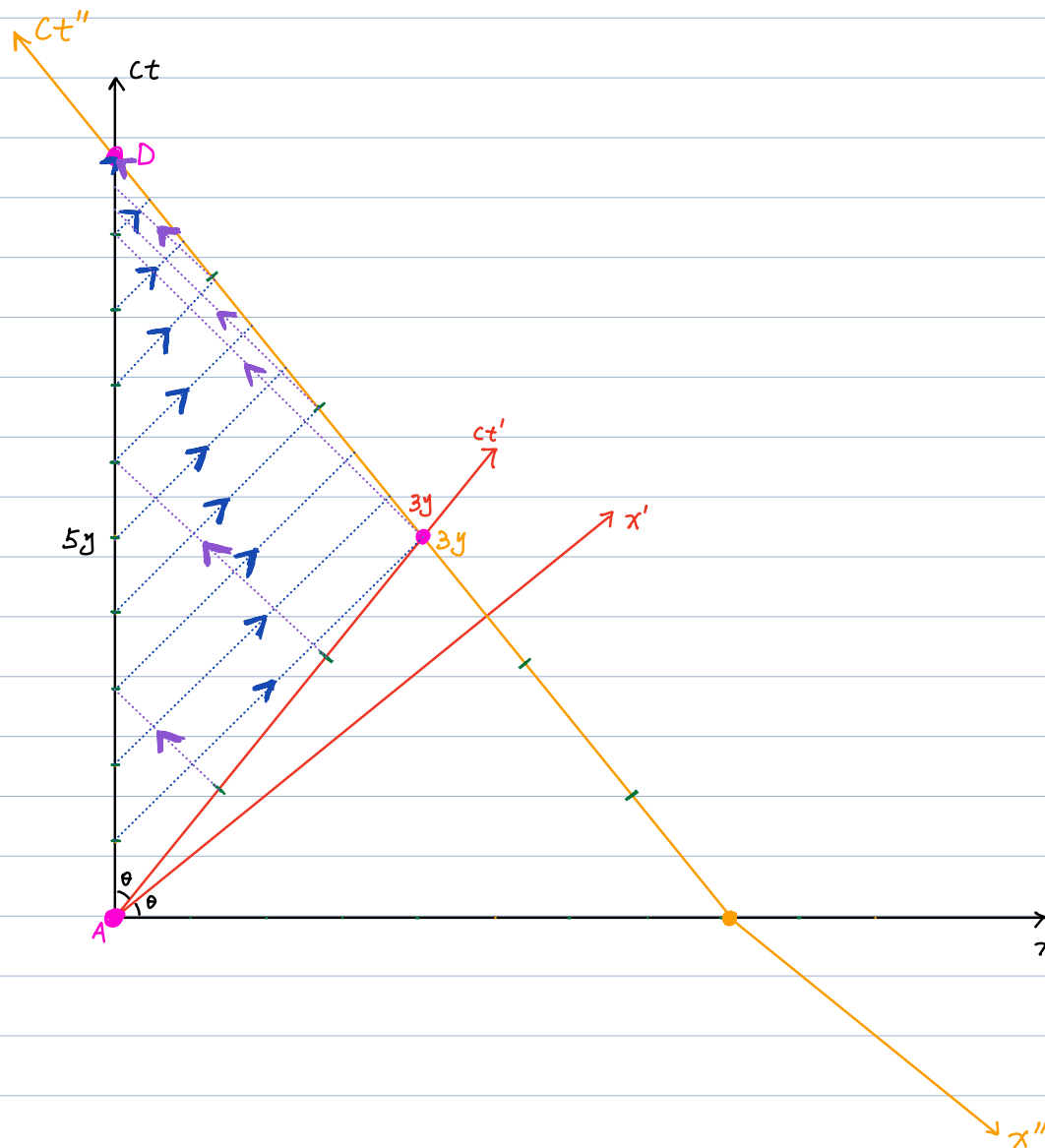
Event **B** in K'' frame:

$$\begin{cases} t_B'' = 3y & (\text{Mary continues counting time}) \\ \chi_B'' = 0 \end{cases}$$

Event **D**: Frank & Mary meet again.

Frank & Mary send radio signals every year

Note: speed of light $\leftrightarrow 45^\circ$ in Minkowski diagram



- Frank sent 10 signals (1 signal per Earth year), aged 10
- Mary sent 6 signals (1 signal per spacecraft year), aged 6
- Doppler effect (signals received per year)